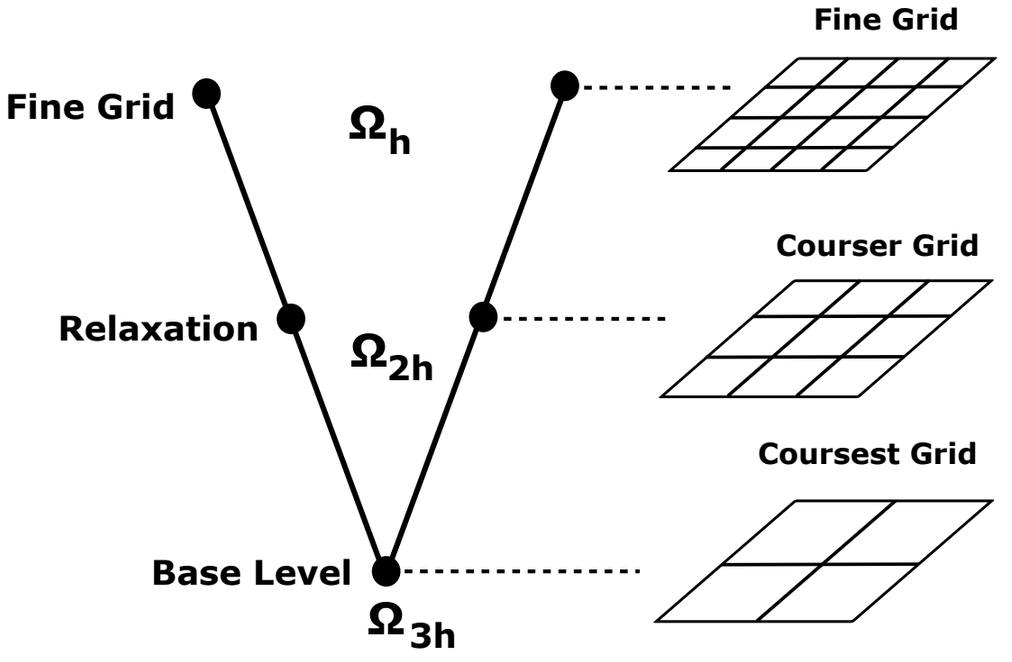
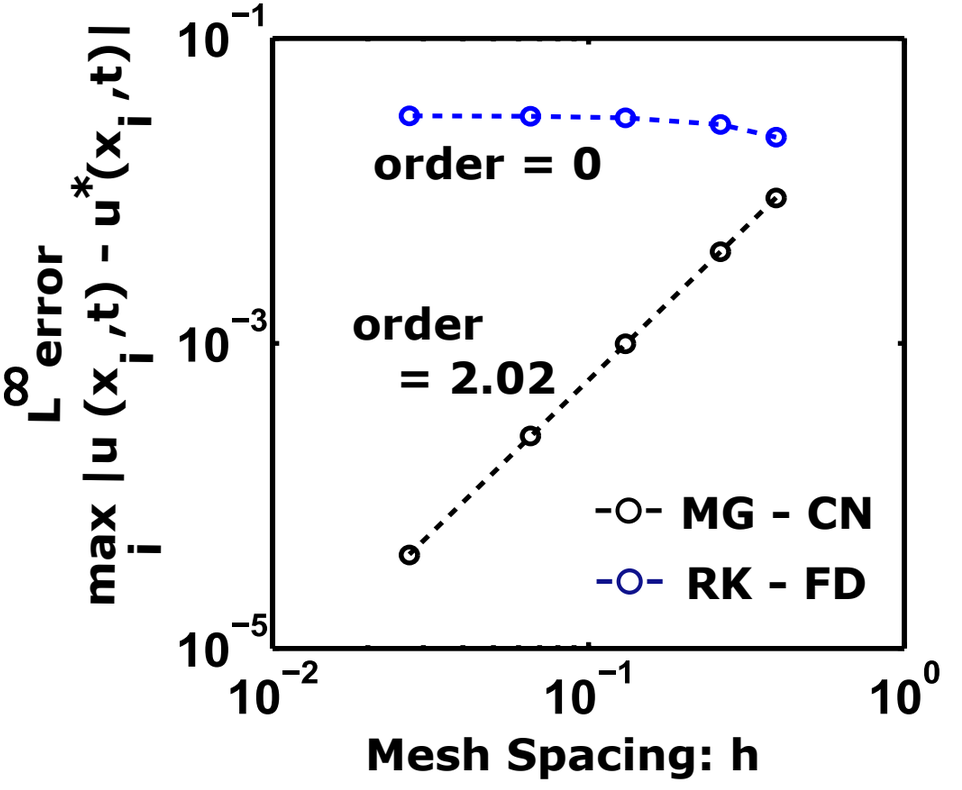
Multigrid Method

(A) Integration: V - cycles



(B) Order of Accuracy



(C) Comparison of Solutions

MG - CN

Δt	L^{∞} error	Order	CPU time
4.0E-2	3.5E-4	1.40	0.07
2.0E-2	2.1E-5	2.24	0.07
1.0E-2	3.6E-5	2.06	0.07
5.0E-3	4.1E-5	2.02	0.07

RK - FD

Δt	L^{∞} error	Order	CPU time
1.0E-3	--	--	--
2.0E-4	3.0E-2	0	2.9E-4
1.0E-4	3.0E-2	0	3.7E-4
5.0E-5	3.0E-2	0	3.0E-4